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AUTOMATED VALVE ACTUATION FOR FRAC FLUID ENDS USING POSITION AND/OR PRESSURE SENSORS

Abstract

A pump includes a power end and a fluid end coupled to the power end. The power end includes a plunger, and the fluid end includes a crossbore in selective fluid communication with inlet and outlet bores. A suction valve controls fluid communication between the inlet bore and the crossbore, and a discharge valve controls fluid communication between the outlet bore and the crossbore. An actuation device is used to move the discharge valve between an open position and a closed position. A position sensor measures a position of the plunger. A pressure sensor measures a pressure of the crossbore, the inlet bore, or the outlet bore. A controller operates the discharge valve based on data from at least one of the position sensor or the pressure sensor.

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Background/Summary

BACKGROUND

Field

[0001] Embodiments of the disclosure relate to a pump, more specifically, to a valve actuation system for fracturing pumps and other pump designs.

Description of the Related Art

[0002] In drilling and completion of a well, fluids are pumped into a wellbore to conduct various downhole operations. For example, fracturing fluids can be pumped downhole to fracture the formation to increase permeability of the formation. In another example, cement can be pumped into an annulus between a wellbore casing and a subterranean surface. After the cement is set, the cement can support and protect the wellbore casing from exterior corrosion and pressure changes.

[0003] Pumps for pumping these fluids typically have a power end and a fluid end. The fluid end may use valves for controlling the flow of fluid in and out of the fluid end. In one example, the valve is operated by a spring. The spring opens when a pressure differential across the valve is sufficient to overcome the biasing force of the spring. However, problems may arise when the valve does not fully open due to a low pressure differential. A partially opened valve will cause the velocity of the fluid to increase, thereby causing more wear on the valve or the valve seat. A low pressure differential may also cause the valve to open and close more frequently, which also leads to wear on the valve or the valve seat.

[0004] There is, therefore, is a need for improved valve designs and valve actuation systems for pumps.

SUMMARY

[0005] In one embodiment, a pump includes a power end and a fluid end coupled to the power end. The power end includes a plunger movable between a retracted position and an extended position. The fluid end includes an inlet bore; an outlet bore; and a crossbore in selective fluid communication with the inlet bore and the outlet bore. A suction valve is provided to control fluid communication between the inlet bore and the crossbore, and a discharge valve is provided to control fluid communication between the outlet bore and the crossbore. An actuation device is used to move the discharge valve between an open position and a closed position. The pump also includes at least one of a position sensor for measuring a position of the plunger or a pressure sensor for measuring a pressure of the crossbore, the inlet bore, or the outlet bore. The pump includes a controller for operating the discharge valve based on data from at least one of the position sensor or the pressure sensor.

[0006] In one embodiment, a method of operating a pump is provided herein. The method includes extending a plunger into a crossbore of the pump and determining a position of the plunger. In response to the plunger crossing a discharge position threshold, a discharge valve in the pump is opened to discharge fluid from the crossbore into an outlet bore of the pump. The method also includes retracting the plunger and determining the position of the plunger. In response to the plunger crossing a suction position threshold, the discharge valve is closed. An inlet bore is opened to allow fluid to flow into the crossbore.

[0007] In one embodiment, a method of operating a pump includes extending a plunger into a crossbore of the pump. The crossbore is in selective fluid communication with an outlet bore and an inlet bore. The method also includes determining a pressure in at least one of the crossbore, the outlet bore, and the inlet bore. In response to the pressure reaching a discharge pressure threshold, a discharge valve in the pump is opened to discharge fluid from the crossbore into the outlet bore. The method also includes retracting the plunger. In response to the pressure reaching a suction pressure threshold, the discharge valve is closed. The inlet bore is opened to allow fluid to flow into the crossbore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0009] FIG. **1** illustrates a pump system, according to one embodiment.

[0010] FIG. **2**A-**2**B illustrate sectional views of a pump from the pump system in FIG. **1** at different positions during operation, according to one embodiment.

[0011] FIG. **3** illustrates an exemplary embodiment of a discharge valve for a pump, according to one embodiment.

[0012] FIG. **4** illustrates an exemplary embodiment of a plunger equipped with a pressure sensor for measuring the pressure in the crossbore, according to one embodiment.

[0013] FIG. **5** illustrates another exemplary embodiment of a plunger equipped with a pressure sensor for measuring the pressure in the crossbore, according to one embodiment.

[0014] For clarity, identical reference numerals have been used, where applicable, to designate identical elements that are common between figures. Additionally, elements of one embodiment may be advantageously adapted for utilization in other embodiments described herein.

DETAILED DESCRIPTION

[0015] In one embodiment, a pump includes a power end and a fluid end coupled to the power end. The power end includes a plunger movable between a retracted position and an extended position. The fluid end includes an inlet bore; an outlet bore; and a crossbore in selective fluid communication with the inlet bore and the outlet bore. A suction valve is provided to control fluid communication between the inlet bore and the crossbore, and a discharge valve is provided to control fluid communication between the outlet bore and the crossbore. An actuation device is used to move the discharge valve between an open position and a closed position. The actuation device can advantageously maintain the discharge valve in the open position during discharge and the closed position during suction. The actuation device beneficially limits the impact between the valve body and the valve seat, thereby reducing wear on the discharge valve. The pump also includes at least one of a position sensor for measuring a position of the plunger or a pressure sensor for measuring a pressure of the crossbore, the inlet bore, or the outlet bore. The pump includes a controller for operating the discharge valve based on data from at least one of the position sensor or the pressure sensor.

[0016] FIG. **1** illustrates a pump system **100**, according to one embodiment. The pump system **100** includes a first pump **102***a* and a second pump **102***b*. The first pump **102***a* includes a fluid end **104***a* and a power end **106***a*. The second pump **102***b* includes a fluid end **104***b* and a power end **106***b*. The pumps **102***a*, **102***b* are mounted in a back-to-back configuration on a platform **108**. For example, the platform **108** may be a skid, truck bed, trailer, etc. In the embodiment illustrated in FIG. **1**, the pumps **102***a*, **102***b* are identical. In another embodiment, the pumps **102***a*, **102***b* may be different. The disclosure of the pump system **100** is exemplarily described as pumping fracturing fluids. However, embodiments of the pumps **102***a*, **102***b* may be applied to pumping cement, mud, and other wellbore fluids.

[0017] According to one embodiment, the pumps **102***a*, **102***b* are compact in size so as to permit the two pumps **102***a*, **102***b* to be oriented in the back-to-back configuration. For example, government regulations often provide vehicle width restrictions for operation on public roadways. In some embodiments, the pump system **100** has a total length L that is less than or equal to a roadway width restriction. For example, the pump system **100** has a total length L equal to or less than 102 inches (i.e., roadway length restriction).

[0018] FIGS. **2**A and **2**B illustrate sectional views of the pump **102***a* from the pump system **100** of FIG. **1** at different positions during operation, according to one embodiment. The pump **102***b* may operate in the same manner. The power end **106***a* is shown in a fully retracted position **200** in FIG. **2**A. The power end **106***a* includes a pump housing **202** and a plunger assembly **204**.

[0019] The pump housing **202** defines an interior volume **209**, which includes a fluid end section **201** and a power end section **203**. The fluid end section **201** is coupled to the fluid end **104***a*. The plunger assembly **204** is disposed within the pump housing **202** and reciprocates between the fluid end section **201** and the power end section **203**. The plunger assembly **204** is operable to cycle between a fully extended position **400** shown in FIG. **2**B and the fully retracted position **200** shown in FIG. **2**A when pumping fluid. For example, the plunger assembly **204** may pump fluid, such as fracturing fluids, through the fluid end **104***a* under high pressure into an oil or gas well.

[0020] The power end **106***a* further includes a crankshaft **212** rotatably mounted in the power end section **203** of the pump housing **202**. The crankshaft **212** includes a crankshaft axis **214** about which the crankshaft **212** rotates. The crankshaft **212** is mounted in the power end section **203** with bearings **216**. The crankshaft **212** further includes a journal **218**, which is a shaft portion to which a connecting rod **220** is attached. The connecting rod **220** includes a crankshaft end **222** and a crosshead end **224**. The crankshaft end **222** is coupled to the crankshaft **212**, and the crosshead end **224** is coupled to a crosshead **206**. The crosshead end **224** may be coupled to the crosshead **206** by a wristpin **225**. In one embodiment, the wristpin **225** is disposed in a cavity **314** of the crosshead **206**.

[0021] The plunger assembly **204** may include the crosshead **206**, a plunger **208**, and a sleeve **210** that forms a fluid seal between the plunger **208** and the crosshead **206** as further described below. The crosshead **206** reciprocates within the pump housing **202** along a plurality of rods **226**, such as two, three, four, or more, disposed in the pump housing **202**. The rods **226** are secured in the pump housing **202** by a retainer member **230**. The crosshead **206** includes an elongated body **228** that may be “T” shaped. The crosshead **206** includes openings **302**, **304** for coupling with the rods **226**. In some embodiments, a bushing is disposed in the openings **302**, **304**, and the bushing encircles and moves along the rods **226**. The elongated body **228** of the crosshead **206** allows for more space within the pump housing **202**. The additional space created in the pump housing **202** by the elongated body **228** of the crosshead **206** allows the sleeve **210** to fit within the pump housing **202**. The elongated body **228** also allows for a longer plunger **208** to be implemented in the pump **102***a*.

[0022] The sleeve **210** is coupled to the crosshead **206** and is at least partially disposed within a space defined between the rods **226**. In one embodiment, the sleeve **210** and the crosshead **206** may be formed as an integral, single piece. In the embodiment shown in FIGS. **2**A and **2**B, the sleeve **210** and the crosshead **206** are shown as separate pieces. The sleeve **210** at least partially surrounds the plunger **208** and an end of the crosshead **206**, forming a fluid seal between the plunger **208** and the crosshead **206**. The plunger **208** is optionally coupled to the crosshead **206** via a retainer plate **234** and screw **236**. The retainer plate **234** sits in a step **238** formed in the sleeve **210**. The screw **236** is disposed through the retainer plate **234** and is threaded into the crosshead **206** at one end and the plunger **208** at an opposite end. In an alternative embodiment, the retainer plate **234** may be coupled directly to the crosshead **206** (such as by cap screws), and the screw **236** may be threaded into the retainer plate **234** at one end and the plunger **208** at an opposite end (as further described below and shown with respect to FIG. **3**B).

[0023] A plurality of sealing members **240** is coupled to the sleeve **210** and the rods **226**. The sleeve **210**, the plurality of sealing members **240**, and the retainer member **230** form a barrier, such as a fluid seal, between the fluid end section **201** and the power end section **203** of the pump housing **202** to prevent fluid contamination between the fluid end section **201** and the power end section **203**. The sleeve **210** moves with the plunger **208** as the crankshaft **212** rotates, thus maintaining the barrier between the power end section **203** and the fluid end section **201** at all points during operation of the crankshaft **212**.

[0024] The fluid end **104***a* is coupled to the fluid end section **201** of the pump housing **202**. The fluid end **104***a* includes a suction valve **290** and a discharge valve **292**. The suction valve **290** is disposed in an inlet bore **242**, and the discharge valve **292** is disposed in an outlet bore **243**. A crossbore **244** is disposed between the inlet bore **242** and the outlet bore **243**, and the crossbore **244** is in selective fluid communication with the inlet bore **242** and the outlet bore **243**. A fluid end seal assembly **246** is disposed adjacent to one end of the crossbore **244** that leads into the fluid end section **201** of the pump housing **202**. In one embodiment, the fluid end seal assembly **246** may include one or more seals, such as an O-ring, positioned within the crossbore **244** to form a fluid seal between the crossbore **244**, the plunger **208**, and the fluid end section **201** of the pump housing **202**. The fluid end seal assembly **246** prevents fluid in the crossbore **244** from flowing into the fluid end section **201** of the pump housing **202**.

[0025] The discharge valve **292** is disposed in the outlet bore **243** to control fluid communication between the outlet bore **243** and the crossbore **244**. The suction valve **290** is disposed in the inlet bore **242** to control fluid communication between the inlet bore **242** and the crossbore **244**. The suction valve **290** includes the same features as the discharge valve **292**, and the discussion herein with respect to the discharge valve **292** is equally applicable to the suction valve **290**.

[0026] FIG. **3** is an exemplary embodiment of the discharge valve **292**. In one embodiment, the discharge valve **292** includes a valve body **320** configured for sealing engagement with a valve seat **340**. The valve seat **340** includes a cylindrical body **342** having a bore **344** extending therethrough. The cylindrical body **342** is disposed in the outlet bore **243** and includes an outer shoulder **348** disposed against a ledge formed in the outlet bore **243**. A sealing member **346**, such as an o-ring, is disposed between the cylindrical body **342** and the outlet bore **243** to prevent fluid communication therebetween. The cylindrical body **342** includes a sealing surface **350** for sealing engagement with the valve body **320**. In one embodiment, the sealing surface **350** tapers radially inward and downwardly for engaging the valve body **320**.

[0027] The valve body **320** includes a seal portion **325** connected to a guide portion **330**. The guide portion **330** is configured for engagement with the inner surface of the valve seat **340**. In one example, the guide portion **330** includes a plurality of guide arms **331** in contact with the inner surface of the valve seat **340**. The plurality of guide arms **331** facilitate alignment of the valve body **320** with respect to the valve seat **340** as the valve body **320** moves toward and away from the valve seat **340**. The plurality of guide arms **331** may be circumferentially spaced and include any suitable number of arms **331**, such as three, four, five or more arms **331**. The seal portion **325** has an outer diameter that is larger than the diameter of the bore **344** of the valve seat **340**. A recess **328** is formed in the outer surface of the seal portion **325** for receiving a valve seal **338**. The valve seal **338** includes a sealing surface **339** for sealing engagement with the sealing surface **350** of the valve seat **340**. In this embodiment, the seal portion **325** includes an optional lower sealing surface **337** adjacent the valve seal **338** for sealing engagement with the sealing surface of the valve seat **340**. In some embodiments, the valve seal **338** is optional, and the lower sealing surface **337** may be extended to engage the sealing surface **350** of the valve seat **340**. In some embodiments, the valve seal **338** is integral with the seal portion **325**. In some embodiments, the valve seal **338** may be sufficiently sized to sealingly engage with the valve seat **340** such that the lower sealing surface **337** is optional.

[0028] In one embodiment, the discharge valve **292** includes a piston and cylinder assembly **370** for moving the valve body **320** between an open position and a closed position. FIG. **3** shows the valve body **320** in the closed position in which the valve seal **338** and the lower sealing surface **337** are engaged with the valve seat **340**. The cylinder **375** of the piston and cylinder assembly **370** is at least partially disposed in a retaining member **307**, such as a retaining nut, and coupled to a piston support **373**. The retaining member **307** is disposed in an access opening **308** of the fluid end **104***a*, and the piston support **373** is disposed inward of the retaining member **307** in the access opening **308**. A sealing member **371** such as an o-ring is disposed between piston support **373** and the access opening **308**. Although the retaining member **307** and the piston support **373** are shown as separate components, it is contemplated that they may be integrated as a single component. The piston support **373** includes a recess **374** for receiving the cylinder **375** and a bore **376** to support the piston **380**. The proximal end of the piston **380** is disposed in a chamber **377** of the cylinder **375**, and the distal end of the piston **380** is attached to the valve body **320**. The proximal end of the piston **380** is movable in the chamber **377** as the piston **380** moves between a retracted position and an extended position. FIG. **3** shows the piston **380** in the extended position, which moves the valve body **320** to the closed position. The piston and cylinder assembly **370** may be powered using hydraulic fluid, pneumatic fluid, or electricity. In addition to a piston and cylinder assembly **370**, it is contemplated other suitable actuation devices may be used to operate the discharge valve **292**. For example, an electric motor or an electric slider may be used to operate the discharge valve **292**. In some embodiments, an optional biasing member **318** is used to bias the valve body **320** toward the valve seat **340**. FIG. **3** shows the biasing member **318** disposed between the valve body **320** and the piston support **373**.

[0029] In some embodiments, the suction valve **290** is operated by the biasing member **318**, as shown in FIGS. **2**A and **2**B. In this example, the piston and cylinder assembly **370** is optional. In some embodiments, at least one of the suction valve **290** and the discharge valve **292** is operated by an actuation device such as the piston and cylinder assembly **370**.

[0030] In some embodiments, operation of at least one of the suction valve **290** and the discharge valve **292** is based on a position of the plunger **208**. In one embodiment, one or more position sensors may be used to determine the position of the plunger **208**. In one example, a position sensor **411**, such as a linear position transducer, is located in the fluid end section **201** of the pump housing **202** to measure the position of the plunger **208**. In another example, a position sensor **412**, such as a linear position transducer, is located adjacent the rod **226** to measure the position of the crosshead **206**, which corresponds to the position of the plunger **208**. In yet another example, a position sensor **413**, such as an encoder, is located adjacent the crankshaft **212** to measure the position of the crankshaft **212**, which corresponds to the position of the plunger **208**.

[0031] The position of the plunger **208** is sent to a controller **420** configured to operate the piston and cylinder assembly **370** to open or close the discharge valve **292** based on the position of the plunger **208**. For example, when the plunger **208** reaches or crosses a discharge position threshold during its extension into the crossbore **244**, the controller **420** can open the discharge valve **292**. In one example, the discharge position threshold is at the beginning of the plunger's **208** extension. In another example, the discharge position threshold is a distance from 1% to 30% of the total distance of the plunger's **208** extension. The controller **420** can close the discharge valve **292** when the plunger **208** reaches or crosses a suction position threshold during its retraction away from the crossbore **244**. In one example, the suction position threshold is when the plunger **208** is at the beginning of its retraction. In another example, the suction position threshold is a distance from 1% to 30% of the total distance of the plunger's **208** retraction.

[0032] In operation, the plunger assembly **204** reciprocates between the power end section **203** and the fluid end section **201** of the pump housing **202**. The plunger **208** of the plunger assembly **204** may extend through the fluid end section **201** of the pump housing **202** and into the crossbore **244** of the fluid end **104***a*. In one embodiment, the plunger assembly **204** has a stroke length of about 6 inches. In another embodiment, the plunger assembly **204** has a stroke length between about 6 inches to 12 inches. In another embodiment, the plunger assembly **204** has a stroke length less than about 6 inches. In yet another embodiment, the plunger assembly **204** has a stroke length greater than about 12 inches.

[0033] The sleeve **210** moves with the plunger **208** as the plunger assembly **204** reciprocates between the power end section **203** and the fluid end section **201**. The sleeve **210** maintains a fluid seal between the power end section **203** and the fluid end section **201** of the pump housing **202** to prevent, during the reciprocating movement of the plunger assembly **204**, cross contamination (of fluids and/or solids) between the fluid end section **201** and the power end section **203**. In one embodiment, the plunger assembly **204** prevents the travel of lubrication fluid from the fluid end section **201** to the power end section **203**, which, over time, may deteriorate and contaminate the power end **106***a* of the pump **102***a*.

[0034] FIG. **2**A shows the plunger **208** in the retracted position and ready for extension. The crossbore **244** is at least partially filled with fluid, and the suction valve **290** is closed due to the pressure in the crossbore **244**. In one example, the discharge position threshold for opening the discharge valve **292** is the beginning of the plunger's **208** extension into the crossbore **244**. The controller **420** will open the discharge valve **292** by retracting the piston **380** of the piston and cylinder assembly **370**. In some examples, the discharge position threshold for opening the discharge valve **292** is a predetermined distance after the start of the plunger's **208** extension. The position of the plunger **208** can be determined using any of the position sensors **411**, **412**, **413**. The controller **420** can delay opening the discharge valve **292** until after the piston **380** has crossed the discharge position threshold to allow the pressure in the crossbore **244** to build, and in some instances, to close the suction valve **290**. As the plunger **208** continues to extend, the piston **380** will maintain the discharge valve **292** in the open position so the fluid in the crossbore **244** can flow into the outlet bore **243**. In this respect, use of the piston and cylinder assembly **370** advantageously prevents chattering of the discharge valve **292** caused by turbulence from the fluid flowing out of the crossbore **244**. In this manner, the impact between the valve body **320** (including the valve seal **338**) and the valve seat **340** can be minimized, thereby reducing wear on the discharge valve **292**. In turn, the costs associated with maintaining or replacing the discharge valve **292** can be reduced.

[0035] After discharging the fluid in the crossbore **244**, the plunger **208** is retracted to refill the crossbore **244**. In one example, the beginning of the plunger's **208** retraction away from the crossbore **244** is the suction position threshold for closing the discharge valve **292**. The controller **420** will close the discharge valve **292** by extending the piston **380** of the piston and cylinder assembly **370** so that the valve seal **338** engages the sealing surface **350** of the valve seat **340**. In some examples, the suction position threshold for closing the discharge valve **292** is a predetermined distance after the start of the retraction. The controller **420** can close the discharge valve **292** after the piston **380** has crossed the suction position threshold. As the plunger **208** retracts, the pressure in the crossbore **244** will decrease, thereby creating a pressure differential between the inlet bore **242** and the crossbore **244** that is sufficient to overcome the biasing force of the biasing member **318** of the suction valve **290**. As a result, the suction valve **290** will open to allow fluid from the inlet bore **242** to enter and at least partially fill the crossbore **244**. During the plunger's retraction, the piston **380** will maintain the discharge valve **292** in the closed position. In this respect, use of the piston and cylinder assembly **370** advantageously prevents chattering of the discharge valve **292** caused by turbulence from the fluid flowing into the crossbore **244**. In this manner, wear on the discharge valve **292** is reduced while the crossbore **244** is being filled. The crossbore **244** will continue to fill until the pressure differential is no longer sufficient to keep the suction valve **290** open. At which time, the process of filling and discharging the crossbore **244** repeats. Although the suction valve **290** is operated by the biasing member **318**, it must be noted that the suction valve **260** can also be operated using an actuation device such as the piston and cylinder assembly **370**.

[0036] In some embodiments, operation of at least one of the suction valve **290** and the discharge valve **292** is based on the pressure in one or more of the crossbore **244**, the outlet bore **243**, and the inlet bore **242**. One or more pressure sensors may be positioned to measure the pressure at one or more of these bores **242**, **243**, **244**. The pressures measured at one or more of these bores **242**, **243**, **244** are sent to the controller **420**, which is configured to operate the piston and cylinder assembly **370** to open or close the discharge valve **292** based on the measured pressures at one or more of these locations.

[0037] In FIG. **4**, the plunger **208** is equipped with a pressure sensor **441** for measuring the pressure in the crossbore **244**. In this example, the pressure sensor **441** is disposed inside the plunger **208** and is configured to measure the pressure in the crossbore **244**. The pressure sensor **441** can be any suitable pressure sensor for measuring the fluid pressure in the crossbore **244**.

[0038] FIG. **5** illustrates an exemplary embodiment of a plunger **508** suitable for use as a plunger in a pump, such as plunger **208** in the pumps **102***a*, **102***b*. In this embodiment, the plunger **508** includes a tubular body **510** having a chamber **520** formed therein. The chamber **520** is open at the front end **516** of the plunger **508** facing the crossbore **244** and is sufficiently sized to house the pressure sensor **441**. The back end **518** of the plunger **508** is configured for coupling with a crosshead, such as crosshead **206**. In this embodiment, the plunger **508** includes a circumferential recess **517** close to the back end **518** to facilitate coupling with the crosshead **206**. The diameter **519** of the back end **518** is larger than the diameter of the recess **517** but smaller than the outer diameter **514** of the plunger **508**.

[0039] A plug **530** is provided to close the front end of the chamber **520**. An optional sealing member **526** is disposed between the plunger **508** and the tubular body **510** defining the chamber **520**. In some embodiments, the outer surface **521** of the plug **530** has a taper for engaging a corresponding taper formed on the inner surface of the tubular body **510**. The plug **530** may be engaged to the tubular body **510** using threads, interference fit, or other suitable engagement mechanisms. A plug bore **532** extends through the plug **530** and provides fluid communication between the crossbore **244** and the pressure sensor **441** in the chamber **520**. The front end of the plug bore **530** is configured to receive a retrieval tool to facilitate removal of the plug **530** from the tubular body **510**. In one example, the front end of the plug bore **530** includes threads **534** for mating with the retrieval tool.

[0040] The chamber **520** is in fluid communication with a plunger bore **522**. The plunger bore **522** is sufficiently sized to accommodate a wire **540** for transferring information, such as measured data, between the pressure sensor **441** and the controller **420**. As shown, the plunger bore **522** exits the tubular **510** at the recess **517**. The plunger bore **522** include an angled section **523** to direct the plunger bore **522** toward the recess **517**. However, it is contemplated the plunger bore **522** may exit the tubular body **510** at the back end **518** or a non-recessed part of the tubular body **510**.

[0041] Referring back to FIG. **4**, the pressure measured by the pressure sensor **441** is sent to the controller **420** for operating the discharge valve **292**, the suction valve **290**, or both. In one example, when the pressure in the crossbore **244** reaches a predetermined discharge pressure threshold, the controller **420** can open the discharge valve **292**. The controller **420** can close the discharge valve **292** when the measured pressure reaches a predetermined suction pressure threshold. Similarly, an outlet pressure sensor **442** can be provided to measure the pressure of the outlet bore **243**. The controller **420** can close the discharge valve **292** when the pressure in the outlet bore **243** reaches a predetermined outlet suction pressure threshold and open the discharge valve **292** when the measured pressure reaches a predetermined outlet discharge pressure threshold. An inlet pressure sensor **443** can be provided to measure the pressure of the inlet bore **242**. The controller **420** can close the discharge valve **292** when the pressure of the inlet bore **242** reaches a predetermined inlet suction pressure threshold and open the discharge valve **292** when the measured pressure reaches a predetermined outlet discharge pressure threshold.

[0042] FIG. **4** shows the plunger **208** in the retracted position and ready for extension. The crossbore **244** is at least partially filled with fluid, and the suction valve **290** is closed due to the pressure in the crossbore **244**. The discharged valve **292** is in the closed position. In one example, the pressure in the crossbore **244** measured by the pressure sensor **441** in the plunger **208** has reached the predetermined discharge pressure threshold. In response, the controller **420** will open the discharge valve **292** by retracting the piston **380** of the piston and cylinder assembly **370**. As the plunger **208** continues to extend, the piston **380** will maintain the discharge valve **292** in the open position so the fluid in the crossbore **244** can flow into the outlet bore **243**. Although the discharge pressure threshold is reached while the plunger **208** is fully retracted, it is contemplated that the discharge pressure threshold may be reached while the plunger **208** is retracting or extending. In the event the discharge pressure threshold is reached during retraction, the controller **420** may optionally open the discharge valve **292** and begin discharging the fluid in the crossbore **244** while the plunger **208** continues to retract. Use of the piston and cylinder assembly **370** to keep the discharge valve **292** in the open position advantageously prevents chattering of the discharge valve **292** caused by turbulence from the fluid flowing out of the crossbore **244**. The controller **420** will continue to monitor the pressure in the crossbore **244** until the suction pressure threshold is reached.

[0043] After discharging the fluid in the crossbore **244**, the plunger **208** is retracted to refill the crossbore **244**. The controller **420** will close the discharge valve **292** when pressure in the crossbore **244** reaches the predetermined suction pressure threshold. The suction pressure threshold may be reached when the plunger **208** is extending, retracting, or fully extended. The controller **420** can close the discharge valve **292** by extending the piston **380** of the piston and cylinder assembly **370** so that the valve seal **338** engages the sealing surface **350** of the valve seat **340**. During retraction, the pressure in the crossbore **244** will decrease, thereby creating a pressure differential between the inlet bore **242** and the crossbore **244** that is sufficient to overcome the biasing force of the biasing member **318** of the suction valve **290**. As a result, the suction valve **290** will open to allow fluid from the inlet bore **242** to enter and at least partially fill the crossbore **244**. The piston **380** will maintain the discharge valve **292** in the closed position until the discharge pressure threshold is reached. In this respect, use of the piston and cylinder assembly **370** advantageously prevents chattering of the discharge valve **292** caused by turbulence from the fluid flowing into the crossbore **244**. In this manner, wear on the discharge valve **292** is reduced while the crossbore **244** is being filled. The crossbore **244** will continue to fill until the pressure differential is no longer sufficient to keep the suction valve **290** open. At which time, the process of filling and discharging the crossbore **244** repeats.

[0044] In some embodiments, the controller **420** can operate the discharge valve **292** based on measured data of at least two of the plunger position, the crossbore pressure, the outlet pressure, and the inlet pressure. In one embodiment, the controller **420** opens or closes the discharge valve **292** based on the plunger's position and one of the measured pressures. In one example, the controller **420** opens the discharge valve **292** after the plunger **208** has reached the discharge position threshold as measured by one of the position sensors **411**, **412**, **413**, and the pressure at the outlet bore **243** (or the crossbore **244** or inlet bore **242**) has reached the discharge pressure threshold as measured by the one of the outlet pressure sensor **442** (or the pressure sensor **441** or inlet pressure sensor **443**). In other words, if only one of the discharge position threshold or the discharge pressure threshold has been reached, the controller **420** will not open the discharge valve **292** until the other threshold has been reached. Similarly, the controller **420** closes the discharge valve **292** after the plunger **208** has reached the suction position threshold as measured by one of the position sensors **411**, **412**, **413**, and the pressure at the outlet bore **243** (or the crossbore **244** or inlet bore **242**) has reached the suction pressure threshold as measured by the one of the outlet pressure sensor **442** (or the pressure sensor **441** or inlet pressure sensor **443**). In other words, if only one of the suction position threshold or the suction pressure threshold has been reached, the controller **420** will not close the discharge valve **292** until the other threshold has been reached.

[0045] In some embodiments, the measured data from the position sensors and the pressure sensors are used to predict when the discharge valve **292** or the suction valve **290** may require maintenance. For example, during operation, the plunger **208** discharge position threshold is initially correlated to a discharge pressure threshold such that when the discharge position threshold is reached, the discharge pressure threshold is also reached or is close to being reached. However, over time, the correlation may begin to weaken such that when the plunger **208** reaches the discharge position threshold, the discharge pressure threshold is not as close to being reached. In this instance, the controller **420** can send a maintenance required signal to the operator.

[0046] In another embodiment, the pumps **102***a*, **106***b* may include a vibration sensor **461** for measuring a vibration of the discharge valve **292** or the suction valve **290**. The controller **420** is configured to receive the vibration data from the vibration sensor **461** and to send out a maintenance signal to the operator. In one example, the controller **420** will send out a maintenance signal if a vibration threshold is measured. In another example, the controller **420** monitors the vibration data to identify a potential trend of wear on the discharge valve **292** or suction valve **290**.

[0047] In some embodiments, the controller **420** is configured to operate the first pump **102***a*, the second pump **102***b*, or both. In some embodiments, the controller **420** is located onsite, such as on the pump **102***a*, or at a remote location. Exemplary pumps **102***a*, **102***b* operable by the controller **420** include a fracturing pump, a cement pump, and a mud pump. Data from the position sensors **411**, **412**, **413**, the pressure sensors **441**, **442**, **443**, and the vibration sensor **461** may be sent to a data storage located onsite or to a remote location. In one example, the measured data is sent to a data van. In another example, the measured data is sent, via satellite, to a cloud system.

[0048] The controller **420** disclosed herein includes a central processing unit (CPU) **423**, a memory **424**, and support circuits **425**. The controller **420** is configured to take action in response to measured data received from the position sensors **411**, **412**, **413**, the pressure sensors **441**, **442**, **443**, and the vibration sensor **461**, including operating the discharge valve **292** and suction valve **290** and sending a maintenance signal. The CPU **423** is a general purpose computer processor configured for use in an industrial setting for monitoring and controlling a pump and operations related thereto. The memory **424** described herein may include random access memory, read only memory, floppy or hard disk drive, or other suitable forms of digital storage, local or remote. The support circuits **425** are conventionally coupled to the CPU **423** and comprise cache, clock circuits, input/output subsystems, power supplies, and the like, and combinations thereof. Software instructions (program) and data can be coded and stored within the memory **424** for instructing a processor within the CPU **423**. A software program (or computer instructions) readable by CPU **423** in the controller **420** determines which action is to be taken in response to information received from the position sensors **411**, **412**, **413**, the pressure sensors **441**, **442**, **443**, and the vibration sensor **461**. Preferably, the program, which is readable by CPU **423** in the controller **420**, includes code, which when executed by the processor (CPU **423**), takes action relating to monitoring and operating the pump described herein. The program will include instructions that are used to control the various hardware and electrical components within the pump to perform the various tasks used to implement the operational schemes described herein.

[0049] In one embodiment, a method of operating a pump includes extending a plunger into a crossbore of the pump. The crossbore is in selective fluid communication with an outlet bore and an inlet bore. The method also includes determining a pressure in at least one of the crossbore, the outlet bore, and the inlet bore. In response to the pressure reaching a discharge pressure threshold, a discharge valve in the pump is opened to discharge fluid from the crossbore into the outlet bore. The method also includes retracting the plunger. In response to the pressure reaching a suction pressure threshold, the discharge valve is closed. The inlet bore is opened to allow fluid to flow into the crossbore.

[0050] In one or more of the embodiments described herein, the method also includes determining a position of the plunger and opening the discharge valve only if the position of the plunger reaches a discharge position threshold.

[0051] In one or more of the embodiments described herein, the method also includes measuring a vibration of one or more of the discharge valve and the suction valve and sending a maintenance required signal to an operator.

[0052] While the foregoing is directed to specific examples, other examples may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A pump, comprising, a power end having a plunger movable between a retracted position and an extended position; and a fluid end coupled to the power end, the fluid end having: an inlet bore; an outlet bore; a crossbore in selective fluid communication with the inlet bore and the outlet bore; a suction valve for controlling fluid communication between the inlet bore and the crossbore; a discharge valve for controlling fluid communication between the outlet bore and the crossbore; a piston and cylinder assembly_for moving the discharge valve between an open position and a closed position, wherein a piston of the piston and cylinder assembly is coupled at one end to a valve body of the discharge valve, and wherein a cylinder of the piston and cylinder assembly is at least partially disposed in a retaining member, which is at least partially disposed in an access opening of the fluid end; at least one of: a position sensor for measuring a position of the plunger; or a pressure sensor for measuring a pressure of the crossbore, the inlet bore, or the outlet bore; and a processing unit controller for operating the discharge valve via the piston and cylinder assembly based on data from at least one of the position sensor or the pressure sensor.

2. The pump of claim 1, wherein the piston and cylinder assembly is powered by hydraulic fluid, pneumatic fluid, or electricity.

3. The pump of claim 1, wherein the position sensor is configured to measure a position of at least one of a crosshead or a crankshaft.

4. The pump of claim 1, further comprising an actuator for moving the suction valve between an open position and a closed position.

5. The pump of claim 1, wherein the pressure sensor is disposed in the plunger and measures the pressure of the crossbore.

6. The pump of claim 5, wherein the plunger includes: a chamber for housing the pressure sensor; and a plug for closing one end of the chamber.

7. (canceled)

8. The pump of claim 1, wherein the piston and cylinder assembly further comprises a support member, wherein the support member includes a bore for supporting the piston, and wherein the support member is disposed inward of the retaining member in the access opening.

9. The pump of claim 1, wherein the processing unit controller operates the discharge valve based on data from the position sensor and the pressure sensor.

10. A pump system, comprising: a first pump having a first fluid end and a first power end; a second pump having a second fluid end and a second power end, wherein the first pump is positioned adjacent the second pump such that the first power end abuts the second power end, wherein the first pump and second pump having a total length less than or equal to 102 inches, and wherein the first pump and the second pump each comprise: a power end having a plunger movable between a retracted position and an extended position; and a fluid end coupled to the power end, the fluid end having: an inlet bore; an outlet bore; a crossbore in selective fluid communication with the inlet bore and the outlet bore; a suction valve for controlling fluid communication between the inlet bore and the crossbore; a discharge valve for controlling fluid communication between the outlet bore and the crossbore; a piston and cylinder assembly for moving the discharge valve between an open position and a closed position, wherein a piston of the piston and cylinder assembly is coupled at one end to a valve body of the discharge valve, and wherein a cylinder of the piston and cylinder assembly is at least partially disposed in a retaining member, which is at least partially disposed in an access opening of the fluid end; at least one of: a position sensor for measuring a position of the plunger; or a pressure sensor for measuring a pressure of the crossbore, the inlet bore, or the outlet bore; and a processing unit controller for operating the discharge valve via the piston and cylinder assembly based on data from at least one of the position sensor or the pressure sensor.

11. The pump system of claim 10, wherein the position sensor is configured to measure the position of the plunger by measuring a position of a crosshead or a position of a crankshaft.

12. The pump system of claim 10, further comprising an actuator for moving the suction valve between an open position and a closed position.

13. The pump system of claim 10, wherein the pressure sensor is disposed in the plunger and measures the pressure of the crossbore.

14. A method of operating a pump, comprising: extending a plunger into a crossbore of the pump; determining a position of the plunger; in response to the plunger crossing a discharge position threshold, actuating a piston and cylinder assembly to open a discharge valve in the pump to discharge fluid from the crossbore into an outlet bore of the pump, wherein a piston of the piston and cylinder assembly is coupled at one end to a valve body of the discharge valve, and wherein a cylinder of the piston and cylinder assembly is at least partially disposed in a retaining member, which is at least partially disposed in an access opening of the pump; retracting the plunger; in response to the plunger crossing a suction position threshold, closing the discharge valve; and opening an inlet bore to allow fluid to flow into the crossbore.

15. The method of claim 14, further comprising: determining a pressure in the crossbore; and opening the discharge valve only if the pressure in the crossbore reaches a discharge pressure threshold.

16. The method claim 14, wherein the piston and cylinder assembly further comprises a support member, wherein the support member includes a bore for supporting the piston, and wherein the support member is disposed inward of the retaining member in the access opening.

17. The method of claim 14, wherein opening the inlet bore comprises opening a suction valve based the inlet bore having a pressure greater than the pressure in the crossbore.

18-20. (canceled)
